

Minguhn Kim

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Education

POSTECH (Pohang University of Science and Technology) Feb. 2025 – Feb. 2027 (Expected)
M.S. in Graduate School of Semiconductor Technology Pohang, South Korea

Laboratory: BEVIL (Broadband Electronics by Virtual Intelligence Lab)

Advisor: Prof. Byungsub Kim

- **Scholarship:** Recipient of the Graduate School Entrance Scholarship.
- **Core Research:** AI-driven EDA: Automated Analog Layout Generation and RL-based Circuit Optimization.
- **Current Focus:** Layout design and automation for an RRAM-based compute-in-memory (CIM) macro.

Chung-Ang University (CAU) Mar. 2019 – Feb. 2025
B.S. in Electrical and Electronic Engineering Seoul, South Korea

- **GPA:** 4.34 / 4.50 (Summa Cum Laude)
- **Military Service:** Sergeant, ROK Army 702nd Special Commando Battalion. (2-years leave of absence)

Key Research & Projects

RRAM-based Compute-In-Memory (CIM) Macro Layout Design & Automation Aug 2026 – Present

Research Contributor

- Completed the custom layout of an SRAM-based RRAM emulation cell for the CIM array.
- Designing and automating the layouts of the RRAM driver circuits (BL, WL, and SL drivers).
- Developed SKILL scripts to automate repetitive Virtuoso layout edits, such as region-based bulk removal of metal shapes and cleanup of pin labels, replacing error-prone manual selection.
- Planning integration with a current-mirror readout stage (replacing a TIA) to deliver voltages to the ADC, fine-tuning layouts through post-layout simulation.

CNN-based Layout Recognition for Generator Parameter Extraction Jul 2026 – Aug 2026
Onboarding Study / Framework & Training Codebase

- Studied a residual CNN framework that classifies cell types and regresses design parameters directly from layout data.
- Analyzed the model architecture and training code, including the use of standard and difference datasets for parameter estimation.

Samsung S.LSI CMOS Image Sensor (CIS) Layout Optimization May 2026 – Jul 2026
Project Collaborator / RL-based Layout Optimization

- Established a GitHub-based collaboration environment to support the research team's multi-contributor workflow.
- Configured an automated training pipeline on the lab server to enable reinforcement learning (RL)-based layout optimization.

A* Router Integration into Layout Generator Jan 2026 – May 2026
Developer / Routing Automation

- Integrated an A* search-based router into a layout generator, enabling automated routing within generated layouts.

Samsung S.LSI AI Cluster Project – DDI SAMP Layout Automation Sep 2025 – Dec 2025
Lead Developer

- Engineered a GUI-based routing automation tool for Display Driver IC (DDI) layout, streamlining industrial design workflows.

- Utilized AI-native development environments (Claude Code) to accelerate software architecture design and algorithm prototyping.

CTLE Optimization via Reinforcement Learning

Feb 2025 – Sep 2025

Research Contributor

- Contributed to a Python-based framework designed to automate Continuous-Time Linear Equalizer (CTLE) parameter tuning.
- Applied Actor-Critic algorithms to bridge the gap between complex analog behaviors and automated design flows.

Technical Skills

- **Custom IC Design:** Cadence Virtuoso (Schematic, Layout), ADE L (Spectre) simulation, pre-/post-layout simulation
- **Physical Verification:** DRC / LVS / PEX (Calibre)
- **Process Technologies:** Samsung 28nm, TSMC 28nm
- **Programming & Automation:** Python (PyTorch, NumPy, Pandas), SKILL (Virtuoso layout scripting), C/C++, Git, Linux
- **AI-assisted Development:** Claude Code

Awards & Languages

- **Dean's Award (College of ICT Engineering):** 2024 CAU Engineering Academic Festival Dec. 2024
- **Academic Excellence Scholarship:** Chung-Ang University (Multiple-semester recipient)
- **Languages:** TOEIC 845 (Jun. 2026), OPIc Advanced Low (AL) (Jan. 2026), NewTEPS 387 (Mar. 2024)

Extracurricular Activities

- **Music Production:** Singer-songwriter with proficiency in piano, guitar, and DAWs; passionate about MIDI composition as a lifelong hobby and aspiration.
- **Physical Fitness:** Maintains disciplined endurance through consistent weight training, running, and swimming.